

Integration Review

Table: On the first page(s) of the book there is a table of integration formulas (mostly from Calc 1). You should know (or be able to quickly derive) the first 18 formulas. Four of the formulas that I would like to highlight (some we have discussed before) are:

$$\begin{aligned}\int \sec(x) dx &= \ln|\sec(x) + \tan(x)| + C & \& \quad \int \tan(x) dx = \ln|\sec(x)| + C \\ \int \csc(x) dx &= \ln|\csc(x) - \cot(x)| + C & \& \quad \int \cot(x) dx = \ln|\sin(x)| + C\end{aligned}$$

When we are evaluating a difficult integral the antiderivative formulas from calculus 1 need to be second nature to us because our main focus will be on applying a technique(s) that we have learned in chapter 3.

General Strategy for Integration:

This process should give you an idea of a general way in which to approach the evaluation of an integral.

1. Simplify the integrand if possible. Possible ways to do this include:
 - Use algebraic manipulation
 - Use a trig identity
2. Look for an obvious substitution (i.e. a standard u -substitution)
3. If steps 1 and 2 have not led to an answer, try to decide which strategy (or strategies) that we have learned in chapter 3 looks like it might be successful. The strategies we have learned (with one new strategy included) are:
 - Integration by parts
 - If what we are integrating consists of only trig functions we may be able to use the methods of 3.2
 - Trigonometric substitution (remember the radical forms that lead to trigonometric substitution)
 - If we have a rational function consider using partial fractions
 - Complete the square
4. Keep trying

Example 1: Find $\int \frac{\sin^3(x)}{\cos(x)} dx$.

Example 2: Evaluate each of the following integrals

(a) $\int \frac{x^3}{\sqrt{1+x^2}} dx$

(b) $\int_0^4 \frac{6z+5}{2z+1} dz$ (Hint: Begin by completing a polynomial long division)

(c)
$$\int \frac{\sin^2(\theta)\cos^2(\theta) + \sin^4(\theta)\cos^2(\theta)}{1 - \sin^4(\theta)} d\theta$$

(d)
$$\int_0^1 \sqrt{\frac{1-x}{1+x}} dx \quad (\text{Hint: Begin by rationalizing the numerator})$$

Example 3: Evaluate each of the following integrals.

(a) $\int \frac{4^x + 10^x}{2^x} dx$ (*Hint: Use Algebra*)

(b) $\int_{-1}^0 \frac{x^2}{x^2 + 1} dx$ (*Hint: Use Algebra*)

(c) $\int \frac{\tan^{-1}(x)}{x^2} dx$